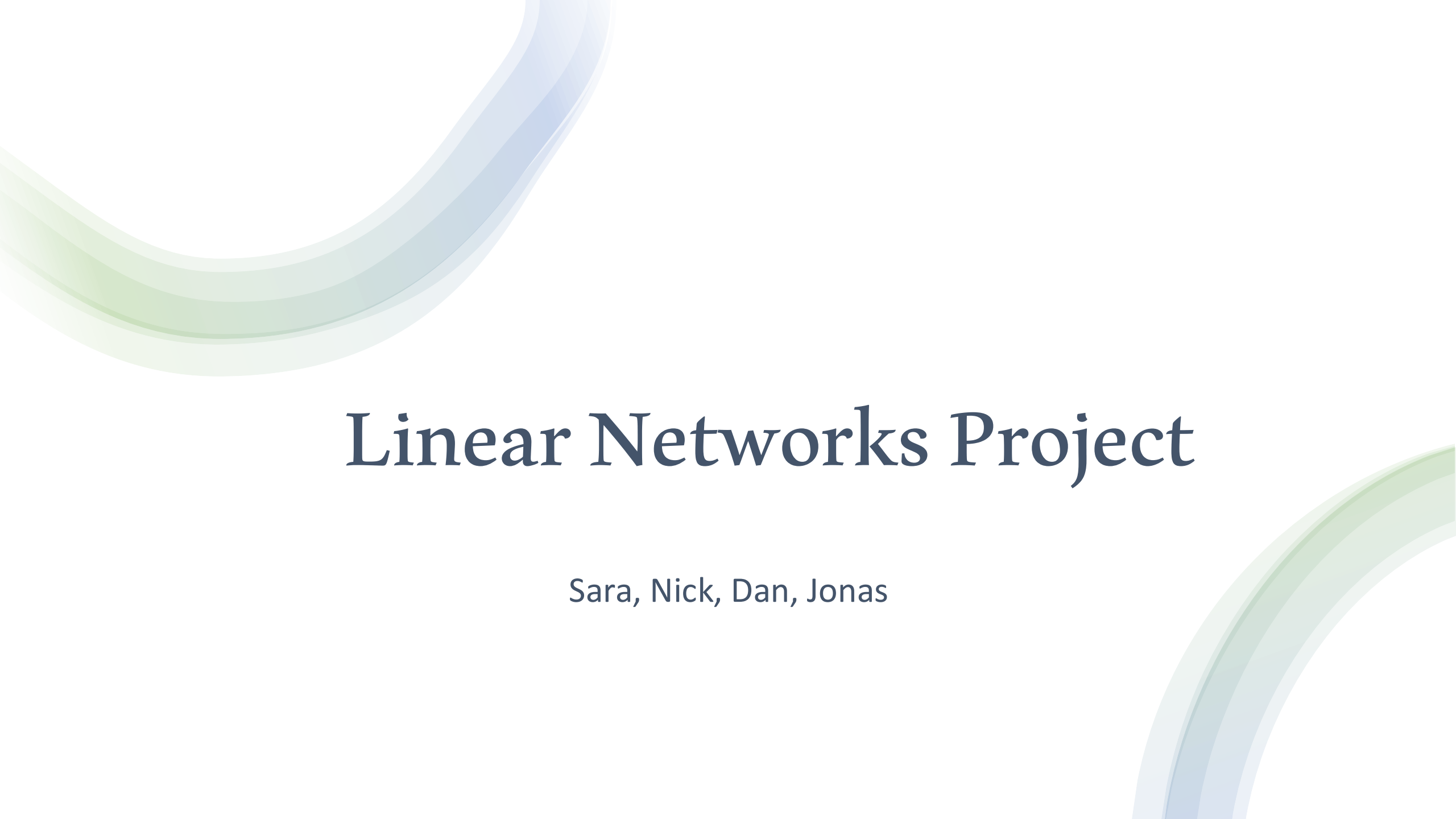


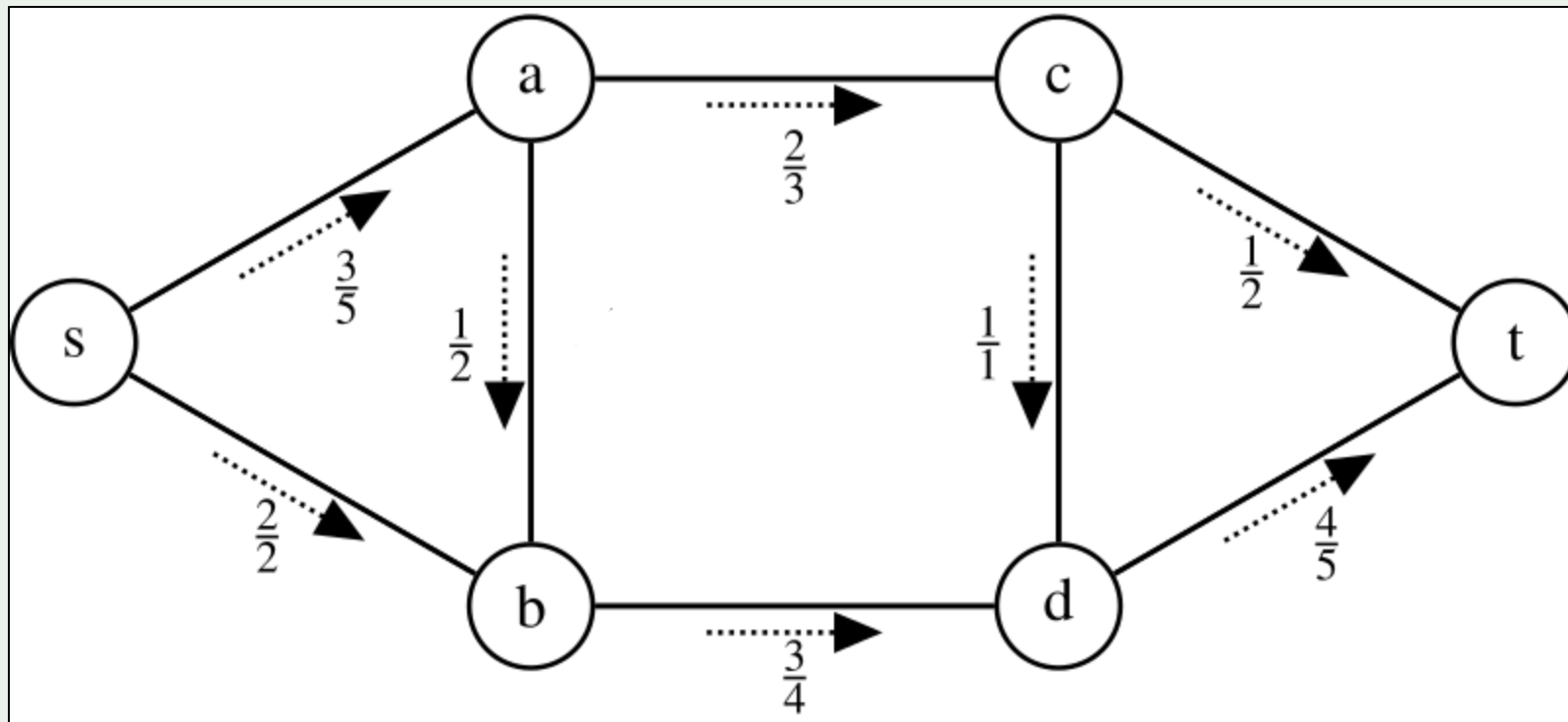
The slide features decorative curved lines in shades of green and blue, positioned in the top-left and bottom-right corners. The main title is centered in a dark blue serif font.

Linear Networks Project

Sara, Nick, Dan, Jonas

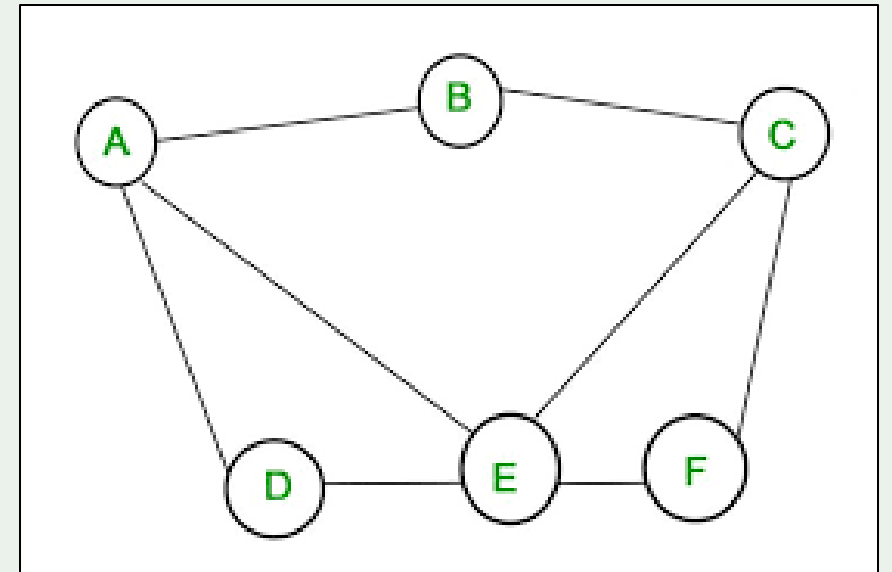
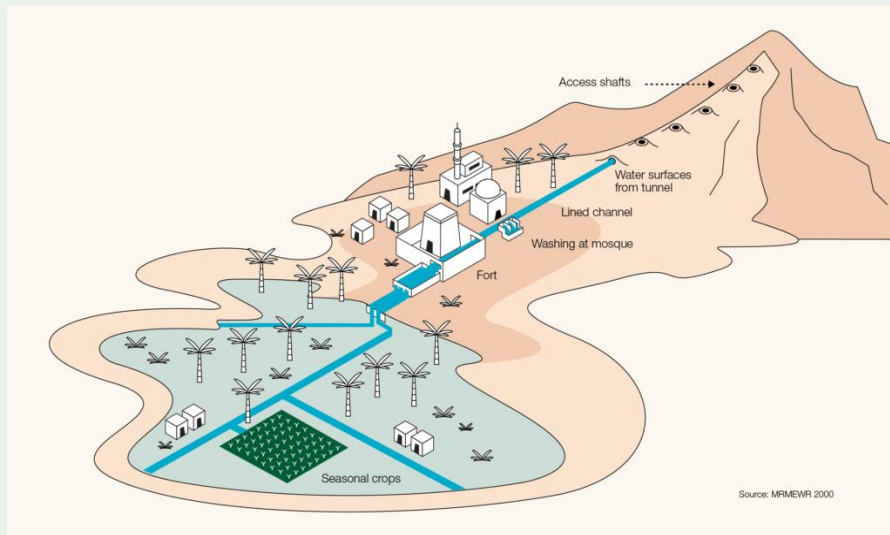
Network Flows

- Directed graphs with capacities and flows on its edges



Before linear algebra

- Network flow was previously described using graphs with nodes and edges before introducing matrix representation.
- Network flows in the world



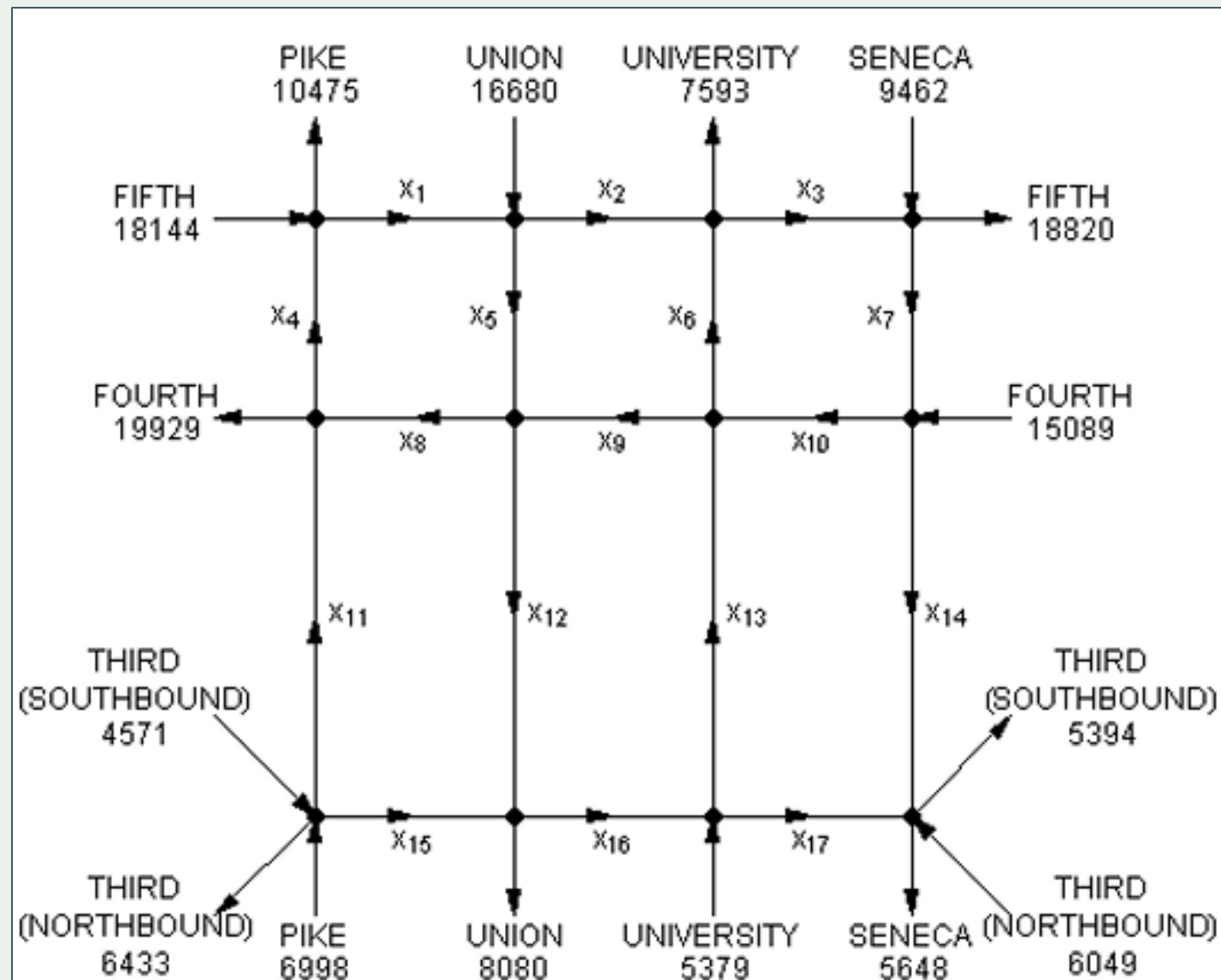
In Linear Algebra

- A network flow problem can be expressed in linear algebra through a matrix
- Each vertex has a row, and all associated input/output edges have a column

"The Big Assumption"

- At each vertex, the flow is 'conserved'; that is, the total input will always be equal to the total output at any given intersection.
- What goes in must come out

Example: Seattle Problem

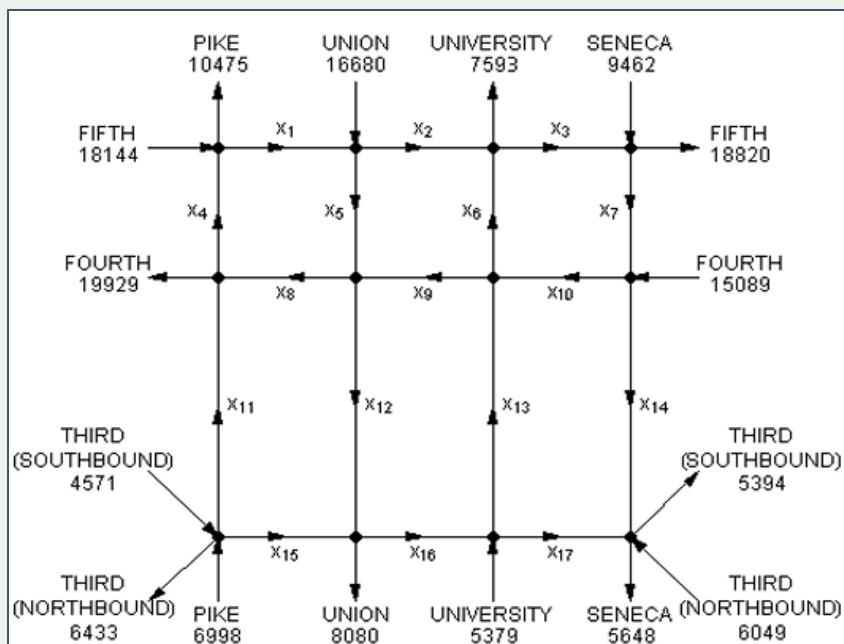


Example: Seattle Problem

In

Out

System of Equations



$$18144 + x_4$$

$$x_1 + 10475$$

$$x_1 - x_2 - x_5 = -16680$$

$$16680 + x_1$$

$$x_2 + x_5$$

$$x_2 + x_6 - x_3 = 7593$$

$$x_2 + x_6$$

$$x_3 + 7593$$

$$x_3 - x_7 = 9358$$

$$9462 + x_3$$

$$18820 + x_7$$

$$-x_{11} - x_8 + x_4 = -19929$$

$$x_{11} + x_8$$

$$x_4 + 19929$$

$$x_5 + x_9 - x_8 - x_{12} = 0$$

$$x_5 + x_9$$

$$x_8 + x_{12}$$

$$x_{10} + x_{13} - x_6 - x_9 = 0$$

$$x_{10} + x_{13}$$

$$x_6 + x_9$$

$$x_7 - x_{10} - x_{14} = -15089$$

$$x_7 + 15089$$

$$x_{10} + x_{14}$$

$$x_{11} + x_{15} = 5136$$

$$4571 + 6998$$

$$x_{11} + x_{15} + 6433$$

$$x_{12} + x_{15} - x_{16} = 8080$$

$$x_{12} + x_{15}$$

$$x_{16} + 8080$$

$$x_{13} + x_{17} - x_{16} = 5379$$

$$x_{16} + 5379$$

$$x_{13} + x_{17}$$

$$x_{17} + x_{14} = 4993$$

$$x_{17} + x_{14} + 6049$$

$$5648 + 5394$$

Example: Seattle Problem

A =

[	1	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	7669;
1	-1	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	-16680;
0	1	-1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	7593;
0	0	1	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	9358;
0	0	0	1	0	0	0	-1	0	0	-1	0	0	0	0	0	0	-19929;
0	0	0	0	1	0	0	-1	1	0	0	-1	0	0	0	0	0	0;
0	0	0	0	0	1	0	0	1	-1	0	0	-1	0	0	0	0	0;
0	0	0	0	0	0	1	0	0	-1	0	0	0	-1	0	0	0	-15089;
0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	5136;
0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	-1	0	8080;
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	-1	1	5379;
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	-1	4993]

$$x_1 - x_4 = 7669$$

$$x_1 - x_2 - x_5 = -16680$$

$$x_2 + x_6 - x_3 = 7593$$

$$x_3 - x_7 = 9358$$

$$-x_{11} - x_8 + x_4 = -19929$$

$$x_5 + x_9 - x_8 - x_{12} = 0$$

$$x_{10} + x_{13} - x_6 - x_9 = 0$$

$$x_7 - x_{10} - x_{14} = -15089$$

$$x_{11} + x_{15} = 5136$$

$$x_{12} + x_{15} - x_{16} = 8080$$

$$x_{13} + x_{17} - x_{16} = 5379$$

$$x_{17} + x_{14} = 4993$$

Example: Seattle Problem

1	0	0	0	0	0	0	-1	0	0	0	0	0	0	1	0	-7124
0	1	0	0	0	0	0	0	-1	0	0	0	0	0	0	10	1474
0	0	1	0	0	0	0	0	0	-1	0	0	0	0	0	0	-738
0	0	0	1	0	0	0	-1	0	0	0	0	0	0	1	0	-14793
0	0	0	0	1	0	0	-1	1	0	0	0	0	0	1	-10	8080
0	0	0	0	0	1	0	0	1	1	0	0	0	0	0	-10	5379
0	0	0	0	0	0	1	0	0	-1	0	0	0	0	0	0	-10096
0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	5136
0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	-10	8080
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	-10	5379
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	4993
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

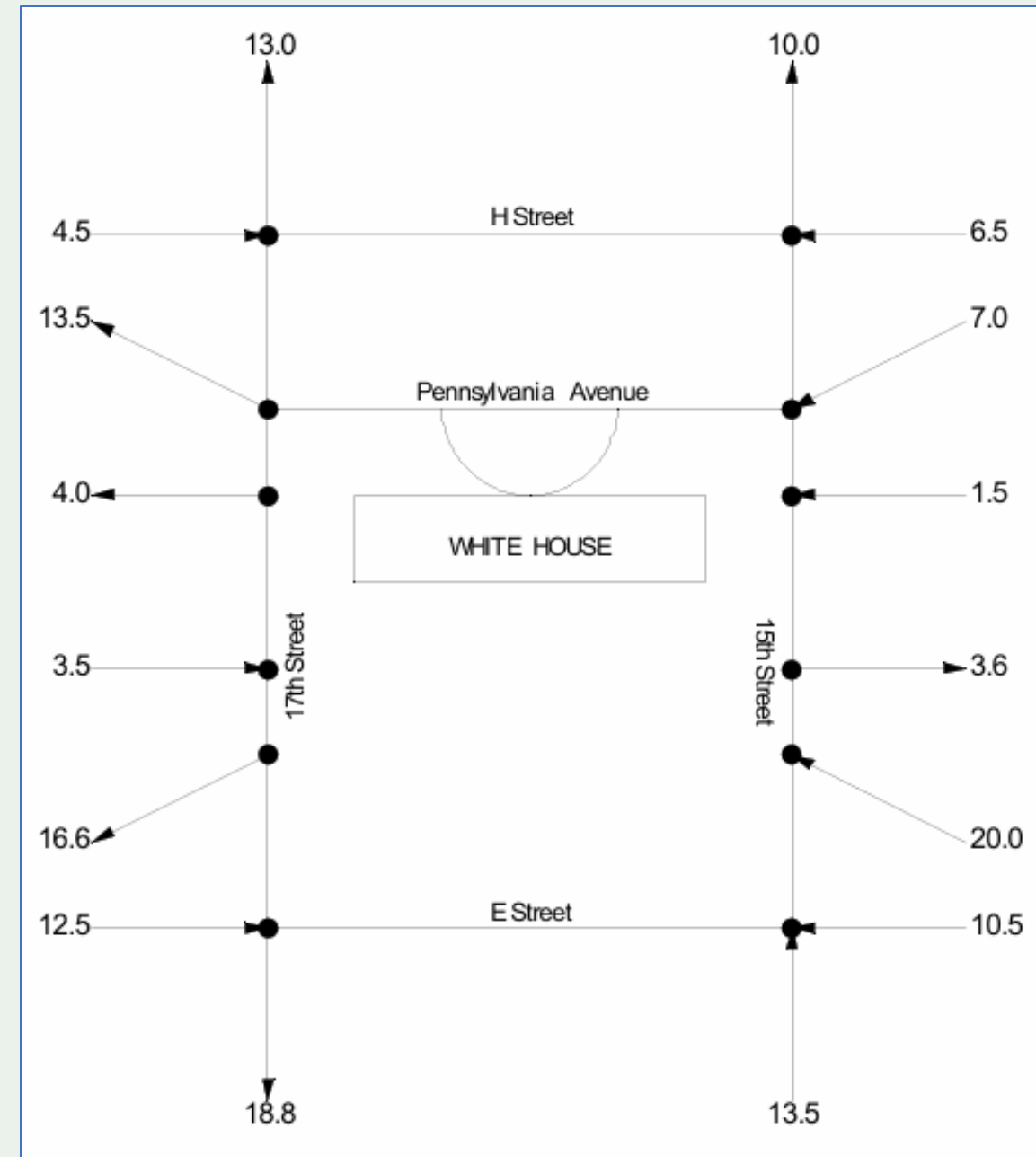
-Why does this matter?

-Possible further applications?

$$\begin{aligned}
 x_1 &= -7124 + x_8 - x_{15} \\
 x_2 &= 1474 + x_9 - x_{16} \\
 x_3 &= -738 + x_{10} \\
 x_4 &= -14793 + x_8 - x_{15} \\
 x_5 &= 8080 + x_8 - x_9 - x_{15} + x_{16} \\
 x_6 &= 5379 - x_9 - x_{10} + x_{16} \\
 x_7 &= -10096 + x_{10} \\
 x_8 &\text{ free} \\
 x_9 &\text{ free} \\
 x_{10} &\text{ free} \\
 x_{11} &= 5136 - x_{15} \\
 x_{12} &= 8080 - x_{15} + x_{16} \\
 x_{13} &= 5379 + x_{16} \\
 x_{14} &= 4993 \\
 x_{15} &\text{ free} \\
 x_{16} &\text{ free} \\
 x_{17} &= 0
 \end{aligned}$$

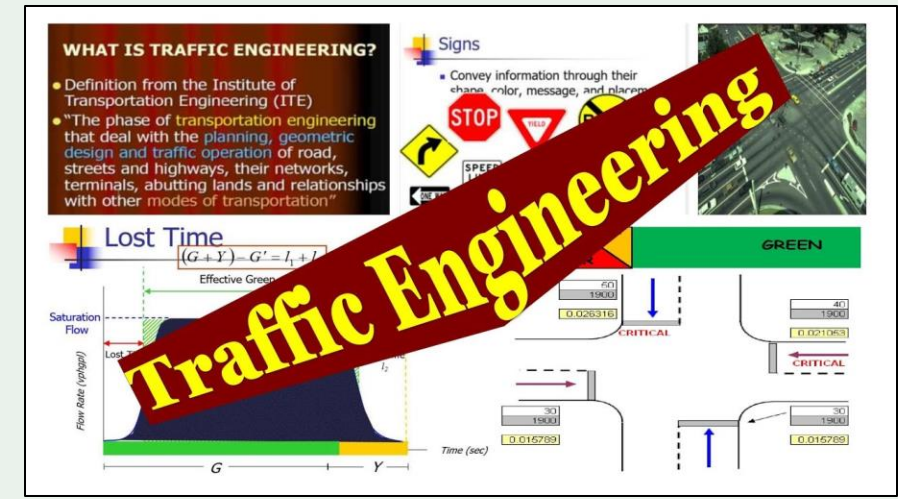
Example: DC Problem

- Bidirectionality (net flows)
- How to tackle this problem?
- What happens during a road closure?



Real world applications

- Civil engineering (roads)
- Computer science (Graph traversal)
- Air traffic flight patterns
- Plumbing (Water flow)
- Electric engineering (Kirchhoff's Current Law)





RADAR VECTORING PROCEDURE AND METHOD

1. Introduction

Air traffic control radars are devices which are used by air traffic controllers to detect, monitor and guide aircraft within a delimited airspace region.

The radar vectoring procedure is one of the management tools which can be used by air traffic controllers (named also ATC) in areas where air traffic control service is provided.

The basic methods of radar vectoring are constituted of several rules in order to handle all aircraft to achieve maximum use of airspace.

Several types of radars are used in the framework of civil or military air traffic control.

At IVAO, we have only one radar system simulated by IvAc software.

2. Condition of radar vectoring use

During increased traffic at the airport, radar vectoring becomes the standard procedure in most of the large airports to arrange aircraft in sequence to ensure air traffic flow management for the most efficient use of airspace.

In some airfield procedures, radar vectoring is a mandatory procedure to guide aircraft on the final approach track.

The radar vectoring procedure shall be used by air traffic controllers only for identified aircraft on the radar system whilst maintaining two way radio communications all the time.

In smaller airports or airports in mountainous areas, the approach controller might provide procedural control (published procedures) only or partially.

2.1. What is radar vectoring

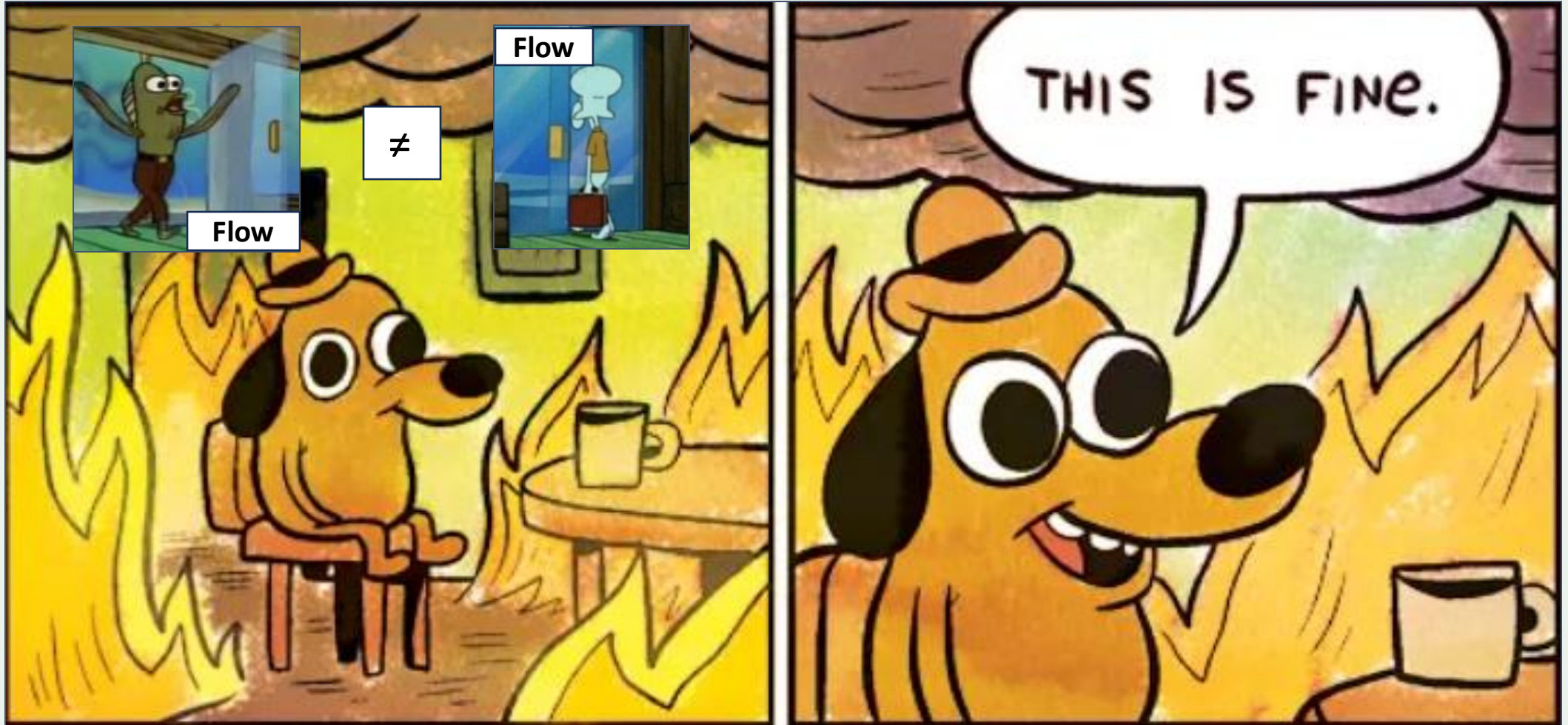
In the radar vectoring procedure, the controller can assign headings, altitudes and speeds to IFR aircraft in order to guide aircraft in his area of responsibility.

An ATC controller shall not take any risk on aircraft safety when using the radar vectoring procedure. An ATC shall not play with the risk of separation loss. He must ensure that a suitable separation is kept every time after he issues a clearance.

Radar vectoring procedure and method	Version 1.1	15 February 2016	Page 1
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This manual is dedicated only for IVAO_® Network activities. This document must not be used in real aviation or in other networks

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References

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